KUMARI PALAK

Software Engineer – Full Stack & Java Backend Development
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PROFESSIONAL SUMMARY

Final-year CSE student (GPA 9.54/10) with 8 months of internship experience at Truminds Software Systems and full-stack experience in Java, Spring Boot, React.js, TypeScript, Node.js, PostgreSQL, and MongoDB. Skilled in REST APIs, database modeling, and concurrent backend systems. Solved 300+ LeetCode problems. Seeking an SDE role.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, C++, SQL
Frontend: React.js, Redux, HTML5, CSS3, Tailwind CSS, Streamlit
Backend: Spring Boot, Node.js, Express.js, REST API design, JWT Authentication, WebSockets, MQTT
Databases: PostgreSQL, MongoDB, MySQL
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Tools & Platforms: Git, GitHub, Postman, Docker, AWS (basics), IntelliJ IDEA, VS Code
ML/AI Exposure: Scikit-learn, TensorFlow, basic Computer Vision

EXPERIENCE

Software Engineer Intern | *Truminds Software Systems Pvt. Ltd., Bengaluru* 04 Dec 2025 – 31 Jul 2026
Project: Event Management System | *React.js, Spring Boot, MongoDB, JWT*

- Built a full-stack event booking platform with separate organizer/attendee flows, seat/slot booking, and history.
- Designed secure Spring Boot REST APIs (JWT auth, event CRUD, bookings) with a MongoDB data model.
- Built a responsive React.js UI with reusable components and form validation via Axios; deployed a tested demo.
- Collaborated with the Engineering team on assigned tasks, following company engineering policies.

PROJECTS

Distributed Job Scheduler | *Java, Spring Boot, React, PostgreSQL, JWT, REST APIs, Docker*

- **Distributed architecture:** Engineered a production-inspired scheduling platform supporting immediate, delayed, scheduled, recurring, and batch jobs behind JWT-secured REST APIs.
- **Concurrency-safe execution:** Eliminated duplicate processing across parallel workers using atomic job claiming with PostgreSQL FOR UPDATE SKIP LOCKED.
- **Fault tolerance:** Implemented worker heartbeats, automatic retries, stale-lease recovery, and a Dead Letter Queue to isolate permanently failing jobs.
- **Observability:** Delivered a Dockerized React dashboard with execution logs and metrics to monitor jobs, workers, queues, and failures.

Afford Medical | *React, TypeScript, REST APIs, Authentication*

- **Priority engine:** Designed a deterministic ranking algorithm scoring notifications by type, keywords, recency, and viewed status, giving consistent, testable ordering.
- **Frontend:** Built a responsive React/TypeScript app with authenticated REST API integration, notification filtering, and viewed-state tracking.
- **Engineering quality:** Created reusable API services with logging middleware, centralized error handling, and local persistence.

Home Automation | *Python, Streamlit, WebSockets, MQTT, DFA*

- **Device control:** Built a Streamlit dashboard to monitor and control simulated Light, Fan, AC, and Door devices.
- **State safety:** Modeled each device as a DFA to block invalid transitions, and added rule-based scheduling for time-triggered automation.
- **Network monitoring:** Simulated MQTT/HTTP communication with connectivity tracking and latency logging, plus QR-based mobile access and live WebSocket latency charts.

AI Placement Prediction System | *Python, Scikit-learn, React*

- **ML pipeline:** Built a classifier estimating placement probability from GPA, backlogs, internships, and DSA practice.
- **Model selection:** Cleaned student data, engineered features, and benchmarked Logistic Regression against Random Forest to choose the best performer.
- **Explainable output:** Developed a React interface showing the predicted probability along with the key factors driving it.

Code for all projects: github.com/Palak1016

EDUCATION

SRM Institute of Science and Technology | *Tiruchirappalli, Tamil Nadu* Expected 2027
B.Tech, Computer Science and Engineering | GPA: 9.54/10

ACHIEVEMENTS

- Solved 300+ DSA problems on LeetCode; maintained a 9.54/10 GPA across six semesters.
- Delivered six full-stack, backend, and AI applications end to end, from schema to working UI.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | AWS Academy Generative AI Foundations | MongoDB Retrieval-Augmented Generation (RAG) | Stanford Algorithms Specialization (Coursera) | Introduction to IoT (NPTEL)